

FACULDADE DE CIÊNCIAS E TECNOLOGIA
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THESIS PROJECT

PhD in Electrical Engineering and Intelligent Systems

Automated Visual Inspection of Electrical Grid Assets using Active Learning and Embedded Vision Systems

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Abstract

The globally growing demand for electrical power, projected to double by 2050, requires extensive upgrades to the Transport and Distribution (T&D) systems. Despite the need for new infrastructures to address this demand, the reliability of ageing electrical T&D systems remains a more critical concern, with power line insulators inspection and replacement representing over 50% of T&D maintenance costs.

To boost inspection efficiency, electrical DSOs are integrating remote inspection technologies into their operations. However, the processing of the growing volume of collected data is currently strongly limited by human interpretation tasks. Automating visual inspections improves efficiency and consistency by addressing the limitations of manual inspections. Manual checks require skilled workers, and are labor-intensive and time-consuming. Human inspectors inconsistent judgments, promoted by fatigue and cognitive bias, can lead to errors and variability in the maintenance quality.

While automated systems help reduce costs and improve consistency, they come with their own expenses and challenges, such as, development and maintenance costs, integration with existing systems efforts, or service failure handling overload. Additionally, achieving performance that exceeds human capabilities remains a challenge. Computer Vision (CV) generalization capabilities have been mostly strengthened by Deep Learning (DL) models. Convolutional Neural Networks (CNNs) enabled early breakthroughs in the field, while Vision Transformers (ViTs) have most recently shown promising improvements in robustness and generalization, particularly in generic CV tasks. Current advancements in Deep Vision (DV) models primarily follow a model-centric approach, emphasizing neural network architecture optimization and improved learning algorithms. However, the critical role of data is often underestimated, leading to developments that may not fully align with industry expectations.

Generic DV models are typically trained on large-scale datasets like ImageNet and COCO, which contain a diverse range of natural scenes and objects. However, these datasets lack the depth needed for specialized industrial domains. Social media, stock photos, and open-source datasets introduce biases, making these models well-suited for consumer applications but less effective for industry-specific tasks. In sectors like manufacturing or healthcare, images from industrial applications often feature repetitive patterns and low-contrast defects that generic models struggle to detect. Additionally, industrial data is often proprietary and confidential, requiring strict security and privacy regulations.

This research aims to develop efficient techniques for automated visual inspection of electrical grid assets, creating new Deep Vision models tailored for this application. The quality of the data, with diverse image modalities, will be guaranteed by human validation and an active learning approach using the expertise of a Distributed System Operator (DSO). To further improve the inspection process, and envisaging real-time operation and edge processing, data collection will be validated at capture time by a CV embedded system, using low complexity, neuromorphic-oriented models. The developed models will be validated in the real-world through the integration into the DSO processes, aiming to evaluate their effectiveness and reliability.

Key-words— Asset Inspection, Electrical Grid, Computer Vision, Active Learning, Embedded Systems, Data-Centric AI

Resumo

O consumo crescente de energia elétrica, que se prevê duplicar até 2050, exige actualizações estensivas nos sistemas de Transporte e Distribuição (T&D). Apesar da necessidade de construção de novas infraestruturas para satisfazer esta procura, a fiabilidade dos sistemas envelhecidos continua a ser uma preocupação mais crítica, com os isoladores de linhas de alta tensão a representarem mais de 50% dos custos de manutenção dos sistemas T&D.

Para aumentar a eficiência das inspeções, os operadores da rede eletrica estão a integrar nas suas operações tecnologias de inspeção remota. No entanto, o processamento do crescente volume de dados recolhidos encontra-se fortemente limitado pelas tarefas de interpretação humana. A automatização das inspeções visuais melhora a eficiência e a consistência, colmatando as limitações das inspeções manuais, que exigem recursos humanos especializados, além de serem morosas e cognitivamente exigentes. Os julgamentos inconsistentes dos inspetores humanos, influenciados pela fadiga e pelo viés cognitivo, podem conduzir a erros e variabilidade.

Embora os sistemas automatizados ajudem a reduzir custos e a melhorar a consistência, também apresentam desafios, como os custos de desenvolvimento e manutenção e a integração com sistemas existentes. Além disso, alcançar um desempenho que supere as capacidades humanas continua a ser um desafio. As capacidades de generalização da visão computacional têm sido amplamente impulsionadas por modelos de aprendizagem profunda. As redes convolucionais possibilitaram avanços iniciais na área, enquanto os transformadores têm demonstrado recentemente melhorias promissoras em robustez e generalização, especialmente em tarefas genéricas. Os avanços atuais em visão profunda seguem principalmente uma abordagem centrada no modelo, focando-se na otimização das arquiteturas de redes neuronais e na melhoria dos algoritmos de aprendizagem. No entanto, o papel crítico dos dados é frequentemente subestimado, levando a desenvolvimentos que podem não corresponder totalmente às expectativas da indústria.

Os modelos genéricos de visão profunda são normalmente treinados em conjuntos de dados de grande escala, como o ImageNet e o COCO, que contêm uma ampla diversidade de cenas e objetos naturais. No entanto, estes conjuntos de dados não possuem a profundidade necessária para domínios industriais especializados. Redes sociais, bancos de imagens e bases de dados abertas introduzem enviesamentos, tornando estes modelos adequados para aplicações de consumo, mas menos eficazes em tarefas específicas da indústria. Nos setores da manufatura e da saúde, por exemplo, as imagens de aplicações industriais apresentam frequentemente padrões repetitivos e defeitos de baixo contraste, que os modelos genéricos têm dificuldade em detetar. Além disso, os dados industriais são frequentemente proprietários e confidenciais, exigindo o cumprimento rigoroso de regulamentos de segurança e privacidade.

Esta investigação visa desenvolver técnicas eficientes para a inspeção visual automatizada de ativos da rede elétrica, criando novos modelos de visão computacional profunda espcializados para esta aplicação. A qualidade dos dados, com diversas modalidades de imagem, será garantida através de validação humana e de uma abordagem de aprendizagem ativa, utilizando a experiência de operadores da rede de distribuição. Para otimizar ainda mais o processo de inspeção, e tendo em vista a operação em tempo real e o processamento na fronteira (edge processing), a recolha de dados será validada no momento da captura por um sistema de visão embebido, utilizando modelos de baixa complexidade orientados para neuromorfismo. Os modelos desenvolvidos serão validados em condições reais através da sua integração nos processos do operador da rede, com o objetivo de avaliar a sua eficácia e fiabilidade.

Palavras-chave— Inspeção de Activos, Rede Eléctrica, Visão Computacional, Aprendizagem Activa, Sistemas Embebidos, IA Centrada em Dados

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Acronyms

AUC Area Under the Curve.

CV Computer Vision.

DL Deep Learning.

DSO Distribution System Operator.

EL Electro-Luminescence.

EPRI Electric Power Research Institute.

GIS Geographical Information Systems.

HV High Voltage.

IEEE Institute of Electrical and Electronics Engineers.

IR Infra-Red.

IRT Infra-Red Thermography.

LiDAR Light Detection And Ranging.

MV Medium Voltage.

NDT Non-Destructive Testing.

NN Neural Network.

O&M Operation and Maintenance.

PL Power Line.

PV Photo-Voltaic.

RF Random Forest.

RGB Red Green Blue.

ROV Remote Operated Vehicle.

SVM Support Vector Machine.

T&D Transmission and Distribution.

UAV Unmanned Aerial Vehicle.

UV Ultra-Violet.

Chapter 1

Introduction

This chapter introduces the broader context of the research problem addressed in this project, outlining the importance of automated visual inspection in electrical grid asset management. It discusses the motivation behind the study, the objectives guiding the research, and the structure of the document, providing an overview of the key topics covered.

1.1 Motivation

The demand for electrical power is growing, driven by increasing consumption and the integration of renewable energy sources. Leading to more complex grid management, requiring significant expansion, and efficient maintenance strategies to ensure reliability. The Institute of Electrical and Electronics Engineers (IEEE) Power and Energy Society emphasizes the importance of advanced asset management and automated techniques for the inspection and maintenance of electrical grid assets, such as insulators and conductors, in its revised IEEE-1808-2024 guide [1].

Electrical Distribution System Operator (DSO) collect visual data during infrastructure inspections to identify flaws and prioritize maintenance decisions. However, the growing volume of data, particularly images, creates a bottleneck in the human interpretation process, as scaling this task through training is time-consuming and costly. To enable widespread adoption of these technologies, further research and development are required to enhance their reliability and applicability in industrial settings.

Automated visual inspection based on computer vision technology offers a promising solution. By integrating deep learning models in the automated inspection workflow, it is possible to speed-up and reduce the cost of the detection of defects and anomalies in grid infrastructure, thus enhancing maintenance and minimizing downtime. While significant advances in Neural Network (NN) and deep learning have shown potential, challenges related to limited data, model scalability, and the need for specialized algorithms remain.

Automating visual inspections improves efficiency and consistency by addressing the limitations of manual inspections, which are labor-intensive, time-consuming, and subject to human error due to fatigue and cognitive bias [2]. Although automated systems help reduce costs and improve consistency, they bring their own challenges, such as high development costs, integration with existing systems, and failure handling [3]. Additionally, achieving performance beyond human capabilities remains a challenge.

1.2 Context

NN have made significant progress thanks to advances in algorithms, data availability, and computational power [4]. This suggests that there is great potential for achieving better results with increasingly complex models, more data, and greater computational resources. However, the current challenge lies in optimizing the use of these three resources to achieve the best possible performance. The rapid development of benefits from abundant textual data [5], achieving human-level performance in Computer Vision (CV) is more difficult. This challenge requires larger datasets, more sophisticated models, and more efficient computing due to the complexity of visual data.

Recent developments in computer vision have been driven by deep learning models, which show a surprising level of robustness to noisy labels, especially with large datasets. However, clean labels consistently result in better performance than noisy ones, highlighting the importance of accurate annotations for model training [6]. Many recent studies have focused on improving neural network architectures or the learning algorithms used in them [7]. In the electrical grid sector, some have focused on asset classification [8, 9], while others have concentrated on defect identification [10–12]. These approaches range from traditional computer vision techniques [13–16], which require less data but more effort in manual feature engineering, to deep learning models [17–19], which require more data and reduce the need for manual feature extraction. As a result, further improvements in model performance may be achieved by either collecting more annotated data or refining the scope of the data through better feature engineering.

Although much of the focus in recent research has been on model-centric approaches, the importance of high-quality data has often been overlooked, leading to progress that does not always align with industry needs. The creation of large datasets is essential for developing automated inspection technologies that can handle the complexities of real-world applications. The Electric Power Research Institute (EPRI) has been addressing this issue by creating and sharing datasets, and by organizing open-source software development competitions [20]. Some strategies to overcome data quantity limitations include data augmentation techniques, such as reflected images [21], or synthetic data generation [22]. However, applying augmentation during the model's pre-processing stages is more effective to ensure that the validation and test datasets remain unique and reliable.

In recent years, transfer learning [23] has been widely used to address the problem of insufficient training data. While it is highly useful for general applications, this approach alone does not solve domain-specific problems. This has led to the rise of a data-centric approach, which focuses on improving data quality rather than just increasing its quantity. In fields with limited datasets, it is crucial to prioritize high-quality data, as even a small set of well-curated examples with accurate annotations can effectively guide neural networks toward achieving the desired learning outcomes [24]. Andrew Ng [25] and his team have been promoting the data-centric AI movement, which suggests that progress can be achieved not by changing neural network architectures but by improving the quality of data. In industries with limited datasets, he emphasizes the importance of focusing on high-quality data. Small well curated datasets with consistent annotations can significantly enhance neural networks' ability to meet their learning objectives [26].

To advance the application of deep learning in computer vision, further research is needed to refine both the theoretical and practical understanding of data quality, particularly in the context of supervised learning for grid asset inspections.

1.3 Objectives

The primary objectives of this research are:

- **Development of Efficient Inspection Models:** Create deep learning-based Computer Vision models designed specifically for electrical grid asset inspection, including the integration of multimodal data sources and domain-specific knowledge.
- **Implementation of Active Learning Strategies:** Investigate active learning techniques to improve data annotation efficiency, reducing the labor-intensive process of manually curating large datasets while ensuring high-quality, expert-validated data.
- **Design of Low-Complexity Embedded Vision Systems:** Develop low-resource models that can be deployed on edge devices, such as UAVs or embedded systems, to enable real-time processing and immediate validation of inspection data.
- **Validation in Real-World Scenarios:** Evaluate the developed models through integration into the DSOs operational workflow to assess their effectiveness in real-world conditions and determine the feasibility of transitioning to automated, AI-driven inspection systems.
- **Dissemination and Industry Collaboration:** Share high-quality datasets, research findings, and practical insights with the industry, contributing to the improvement of grid monitoring and inspection practices.

These objectives are measurable, achievable, and time-bound, with clearly defined stages and milestones, from literature review and model development to real-world testing and integration.

1.4 Document Overview

This document is organized to provide a comprehensive account of the proposed research and its possible contributions to the field of automated visual inspection of electrical grid assets using deep learning. The chapters are structured as follows:

- **Chapter 1: Introduction** – Introduces the background, objectives, and problem statement that the research is intended to addresses.
- **Chapter 2: State of the Art** – Reviews the current research and technologies related to electrical grid asset inspection, focusing on automated visual inspections, active learning techniques, and embedded vision systems.
- **Chapter 3: Proposed Work Plan** – Provides a detailed description of the planned research, including objectives, main tasks, project timeline, and host conditions.
- **Chapter 4: Developed Work** – Outlines initial tasks results and directions, related to the implementation of standard deep learning architectures for visual inspection and the impact of annotation procedures on model performance.
- **Chapter 5: Conclusions** – Summarizes the expected outcomes of the project and discusses potential challenges and alternative approaches.

Chapter 2

Automated Visual Inspection State-of-the-Art

The state-of-the-art chapter presents a review of the current research and technological developments relevant to automated visual inspections of electrical grid assets. It explores the existing methods, tools, and strategies used in the field, highlighting their strengths and limitations. This chapter serves as a foundation for understanding the existing gaps that the proposed research aims to address.

The management of electrical grid assets has traditionally relied on manual inspection, which is labor-intensive and prone to human error. With the increasing complexity of modern grids, automated solutions are necessary to maintain grid reliability. Automated visual inspection systems, such as those utilizing drones and embedded systems, have gained attention as effective tools for defect detection, condition monitoring, and maintenance planning. These systems use sensors (e.g., cameras, infrared, and UV imaging) to capture images and detect issues like corrosion, wear, or overheating.

Recent work in this area has shown that deep learning-based models, particularly convolutional neural networks (CNNs), can be used for asset classification and defect identification. However, these approaches face challenges in data quality, model generalization, and the computational complexity of large-scale deployment.

2.1 Power Generation Electrical Grid Assets

The considered electrical assets used in power generation systems, such as photovoltaic fields, wind turbine blades, and hydroelectric dam walls, share common characteristics that make them ideal targets for automated visual inspections. These assets are often massive in size, with extensive surface areas that cannot be captured in a single shot, requiring a complex acquisition process involving images from various positions, distance and angles. Additionally, the inspection process must include an ID tracking system to accurately correlate the captured images with specific parts of the assets to provide a complete visualization.

2.1.1 Photovoltaic Modules and Cells

Monitoring and maintaining the integrity of solar modules is crucial for ensuring optimal performance in photovoltaic systems. The visual identification of defects poses a challenge, even for trained experts. Not all conditions affecting a module's efficiency are visible, and conversely, visible flaws don't always impact efficiency. To address this, non-destructive methods such as Infra-Red (IR) and Electro-Luminescence (EL) imaging offer promising solutions. The most commonly adopted platform for Photo-Voltaic (PV) plants inspection is a Unmanned Aerial Vehicle (UAV) platform equipped with thermographic cameras. This setup generates Infra-Red Thermography (IRT) data with annotations that may vary according to the task description.

The IEC TS 62446-3 standard [27] specifies the conditions under which IRT PV inspections should be carried out to ensure accurate and reliable results, such as ambient temperature, irradiance levels, and viewing angles. It provides guidelines for interpreting thermal images and identifying different types of faults based on temperature variations and thermal patterns. The standard includes criteria for classifying and prioritizing detected anomalies for maintenance actions. It also specifies the format and content for inspection reports to ensure consistency and comprehensiveness, including detailed descriptions of identified issues. Some approaches address this task from a simpler perspective, focusing solely on identifying hot-spots in PV modules. This can be achieved using techniques [28, 29] such as adaptive thresholding or advanced Deep Learning (DL) methods [30, 31] like object detection or instance segmentation architectures.

Vincenzo Carletti *et al.* [28] relied on to identify hotspots in PV modules from UAV-collected thermal videos. Qiuping Zheng *et al.* [30] created their own annotations for a subset of the videos collected by Carletti *et al.* [28], labeling a total of 125,722 PV modules and 8,070 hot spots in 5,600 frames. This approach has the advantage of straightforward implementation, allowing any frame of a UAV-captured thermal video to be used as input data for processing, either offline or on an embedded system. However, this simplicity comes with limitations. These methods may not generate valuable information for subsequent maintenance procedures, which require specific details such as the precise location of the failure, the short and long-term impact on power restrictions, and root cause analysis along with possible repair procedures.

A more comprehensive approach considers the system's ability to identify a set of anomaly classes, such as the 12 defined in the IEC TS 62446-3 standard [27]. This standard relates thermal patterns to the elements causing each failure, thereby aiding in the diagnosis and repair. Minhhuy Le *et al.* [32] develop a deep learning model based on ensemble techniques for classifiyng PV thermal paterns using a publicly available dataset [33] containing 20,000 IRT images of cropped PV modules at a resolution of 40x24x1. Each image is labeled as one of 12 classes: a healthy PV module or one of 11 different anomalies (Cell, Cell-Multi, Cracking, HotSpot, Hot-Spot-Multi, Shadowing, Diode, Diode-Multi, Vegetation, Soiling, Offline-Module). In another

study, Minhhuy Le *et al.* [34] proposed a CNN-PVE model based on custom CNN for an embedded system. They performed comparisons against Support Vector Machine (SVM), Random Forest (RF), and applied transfer learning on VGG16, ResNet50, and MobileNet models. However, during a qualitative evaluation of the results, the authors identified ground truth labels that were not aligned with the thermal pattern descriptions provided in the dataset instructions. This discrepancy might lead to question the quantity of noisy annotations in the dataset and their impact on the developed model.

Lukas Bommes *et al.* [35] gathered a dataset of 231,172 frames containing 8 hours of thermographic video footage from seven large PV plants using a UAV equipped with . They annotated segmentation masks and bounding boxes for 26,612 PV modules in 862 video frames, using 93% of the frames for fine-tuning a Mask R-CNN model and 7% for validating it, achieving an AP at an IoU threshold of 0.5 of 99.55%. In another publication, L. Bommes *et al.* [36] framed fault detection as a more realistic unsupervised domain adaptation problem. They recognized that related works typically sample training and testing data from the same distribution, ignoring the presence of domain shifts between data from different PV plants collected under various environmental conditions. To address this, they utilized labeled data from one source PV plant to make predictions on unlabeled data from another target plant. They trained a ResNet-34 convolutional neural network with a supervised contrastive loss to generate low-dimensional embeddings. These embeddings were then input into a k-nearest neighbor classifier to detect anomalies in unseen data, achieving Area Under the Curve (AUC) scores ranging from 73.3% to 96.6% across combinations of source and target datasets, without the need for hyperparameter adjustments. Identifying another key problem in the integration of fully automated inspections with Operation and Maintenance (O&M) procedures, Bommes *et al.* [37] developed a tool, illustrated in Figure 1, for the extraction and georeferencing of PV modules from aerial IR videos. They overcame the low resolution of the GPS receiver device by interpolating geocoordinates for key frames. Tracking and triangulating each segmented module allowed them to successfully map 99.3% of the PV plant modules.

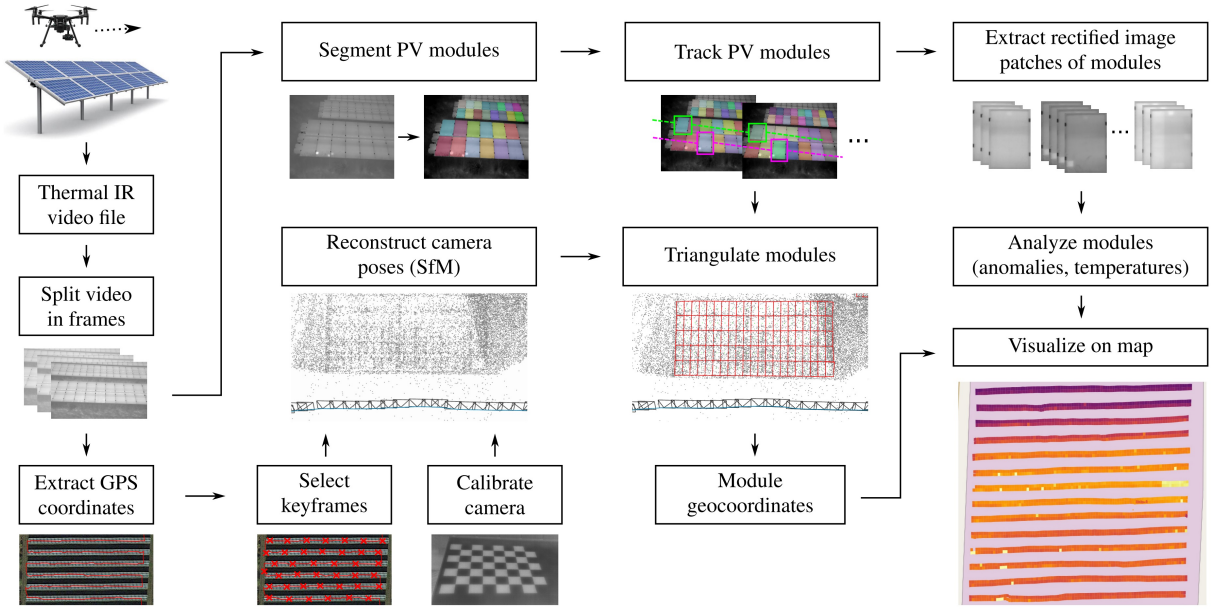
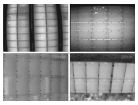
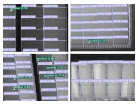
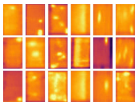
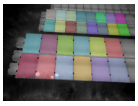
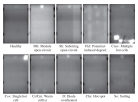


Figure 1: Overview of automatic extraction and georeferencing of PV modules from aerial IR videos [37]

Table 2.1 summarizes the literature details and results on photovoltaic modules using IRT sensors on UAV platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.1: Literature Details on Photovoltaic Modules employing IRT Sensors on UAV Platforms

Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Hot-Spot (HS) Detection in PV Modules		14,215 frames 640 × 480 x 1 336 × 256 x 1 320 × 240 x 1	3,654 Local HS 76 Global HS	Available Upon Request [28]	Explicit DIP	Raindrop	F1 = 77.0%	[28]
Hot-Spot (HS) and PV Modules Detection		5,600 frames 640 × 480 x 1 336 × 256 x 1 320 × 240 x 1	8,070 Local HS 125,722 PV modules	Not Available	Supervised DL	S-YOLOv5	F1 = 96.6%	[30]
PV Modules Thermal Pattern Classification		20,000 IR Patches 40 x 24 x 1	12 Classes of Anomalies	Openly Available [33]	Supervised DL Supervised DL	Ensemble CNN ResNet based Custom CNN 4 Conv2D	F1 = 74,92% F1 = 73.73%	[32] [34]
PV Modules Segmentation		862 frames 640 x 512 x 1	26,612 PV modules	Code and Data Sample Available [38, 39]	Supervised DL + DIP Supervised DL + DIP	Mask R-CNN Mask R-CNN + GPS + SfM	mAP50 = 87.8% mAP50 = 99.3%	[35] [37]
PV Modules Thermal Pattern Classification		4.16 million IR Patches 640 x 512 x 1	12 Classes of Anomalies	Available Upon Request [36]	Supervised DL Semi- -Supervised	ResNet-50 ResNet-34 + kNN Classifier	F1 = 90.68% AuROC from 73.3% to 96.6%	[35] [36]

IR imaging detects damaged cells by capturing the infrared radiation emitted when solar cells stop converting energy into electricity due to disconnection from the circuit. However, the resolution limitations of IR cameras might hinder the detection of minute defects, such as microcracks that haven't yet impacted conversion efficiency. On the other hand, EL imaging offers higher resolution, revealing premature defects. While effective, EL imaging requires trained specialists and manual inspection, making it time-consuming and costly. However, the potential for automated defect classification in EL images presents an opportunity to overcome these challenges.

The principle of electroluminescence involves applying a voltage bias to a PV cell, generating an electric field within the cell. Defects, cracks, or non-uniformities in the silicon material can lead to localized variations in the electric field. When carriers (electrons and holes) recombine in these areas, they emit light, a phenomenon known as electroluminescence. Dark regions or patterns in electroluminescence images may indicate defects, cracks, or other irregularities. EL imaging is employed during the manufacturing process to identify defects early on, ensuring the production of high-quality PV cells. In the field, EL imaging can be used to identify early failures in installed PV panels and prevent a performance drop.

The IEC TS 60904-13 standard [40] provides guidelines for electroluminescence (EL) imaging of PV cells and modules to accurately identify and report defects, aiding in PV system performance maintenance. It details methodologies for capturing and processing EL images, calculating signal-to-noise ratios (SNR), and sets minimum SNR values for various applications (outdoor, industrial, laboratory). The standard also includes criteria for classifying and prioritizing anomalies, specifies the format for inspection reports, and provides guidance on classifying potential cracks and interpreting EL images of crystalline silicon PV modules.

Claudia Buerhop-Lutz *et al.* [41] published an open source dataset [42] that contains 2,624 samples of 300x300 pixels 8-bit grayscale images of functional and defective PV cells with varying degree of degradation, extracted from 44 different PV modules, more precisely, 26 monocrystalline and 18 polycrystalline. Given the unavailability of actual measurements for power degradation,

a pseudo target variable was generated based on the assessment made by electroluminescence experts and considering their confidence levels in each cell image. A label of 1.0 was assigned when the expert was highly confident that the cell image contained a defect. For evaluations of defects with lower confidence, a label of 0.667 was utilized. Functional cells assessed with high confidence were labeled as 0.0, while those assessed with less confidence received a label of 0.333.

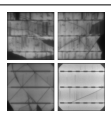
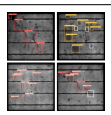
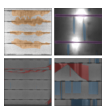
Sergiu Deitsch *et al.*introduced a fully automatic segmentation method [43] designed to precisely extract solar cells from high-resolution EL images. This segmentation method proves robust against underexposure and performs effectively even in the presence of severe defects on solar cells. This robustness can be attributed to the proposed preprocessing techniques and the integration of ridgeness filtering, alongside tensor voting, to accurately determine inter-cell borders and busbars. Notably, the segmentation method exhibits high accuracy (97.80%), enabling its output to be used for subsequent inspection tasks, including automatic classification of defective solar cells and the prediction of potential power loss.

Recognizing the limitations of the ELPV dataset [42], Binyi Su *et al.* [44] introduced PVEL-AD, a dataset of electroluminescence images of photovoltaic cells for anomaly detection. Collected in an industrial manufacturing setup, this dataset contains 36,543 near-infrared images with a resolution of 1024 x 1024 pixels. Huan Fu and Guoqing Cheng [45] used the PVEL-AD dataset [46] to train and test a YOLOv7 model. Joseph Fioresi *et al.* [47] propose automating defect detection and localization in electroluminescence images of photovoltaic cells using semantic segmentation.

Abhishek Jha *et al.* [48] propose overcoming the high labor of detailed annotations and advancing automatic photovoltaic defect detection in electroluminescence images using semi-supervised learning on semantic segmentation. They used the UCF-EL dataset [49] to train both a model and a model, based on a student-teacher methodology.

Table 2.2 summarizes the literature details and results on photovoltaic cells using sensors on fixed platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.2: Literature Details on Photovoltaic Cells employing Sensors on Fixed Platforms

Task Description	**Data**				**Model**			
	Sample	**Size**	**Annotations**	**Source**	**Algorithm**	**Architecture**	**Performance**	**Ref**
PV Cells Classification		2,624 samples 300 x 300 x 1	4 Classes of Pseudo Degradation Level	Openly Available [42]	DIP + ML Supervised Supervised DL	KAZE + SVM CNN VGG19	F1 = 82.52% F1 = 88.39%	[50] [50]
PV Cells Defects Detection		36,543 samples 1024 x 1024 x 1	37,380 bounding box 4 Classes	Available Upon Request [46]	Supervised DL Supervised DL	YOLOv5l ELCN-YOLOv7	mAP50 = 81.93% mAP50 = 91.93%	[44] [45]
PV Cells Defects Segmentation		17,064 samples 165 x 165 250 x 250 400 x 400	9 Classes of Anomalies	Model + Data Openly Available [49]	Supervised DL Semi-Supervised DL	DeepLabv3 w/ ResNet-50 Deeplabv3+	F1 = 69.6% F1 = 84.24%	[35] [48]

2.1.2 Wind Turbine Blades

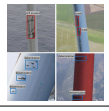
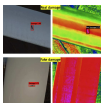
Rotor blades are key components of a wind turbine's power generation system, with their condition significantly affecting overall performance and accounting for a substantial portion of O&M costs.Among the various Non-Destructive Testing (NDT) methods used to inspect surface and sub-surface ,processing visual data collected by UAVs has emerged as a prominent research topic due to its accessibility, safety, and economic advantages,while integrating deep learning algorithms to process visible light images reinforces the symbiotic relationship with these data collection methods.

The EPRI conducted an industry survey to document current practices in categorizing blade damage and defects [51]. The findings help shape a proposed standardized categorization system and reveal gaps in existing practices, such as the need for concrete definitions of categories. This report reflects current industry approaches and offers a foundation for establishing a global standard for damage classification. It defines five categories of damage, ranging from minor to severe, providing examples with photographs and descriptions, outlining system impacts, and recommending actions. This framework is intended to aid in the consistent identification and response to blade damage across the industry.

Shihavuddin *et al.* [52] introduced a high-resolution image dataset [53] for wind turbine surface damage detection, consisting of 558 frames with a resolution of 5280x2970, captured from a single wind turbine tower. Imad Gohar *et al.* [54] investigated the impact of improved annotations on defect detection in wind turbine blade images by creating and sharing [55] 796 annotations across five classes. Cong Yang *et al.* [56] propose a coarse-to-fine image stitching algorithm for efficient wind turbine blade inspection. Syed Zeeshan Rizvi *et al.* [57] improved the efficiency of defect detection on wind turbine blades by shifting the focus toward dataset pre-processing rather than architectural improvements. Weijie Zhou *et al.* [58] addressed the issue of false detection rates by implementing a feature fusion module within a YOLOv7 architecture that utilizes both visible and infrared imagery data.

Table 2.3 summarizes the literature details and results on wind turbine blades using Red Green Blue (RGB) or IR sensors on UAV platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.3: Literature Details on Wind Turbine Blades employing RGB or IR Sensors on UAV Platforms

Task Description	**Data**				**Model**			
	Sample	**Size**	**Annotations**	**Source**	**Algorithm**	**Architecture**	**Performance**	**Ref**
Wind Turbine Surface Defects Detection		558 images 5280 x 2970 x 3	796 bounding box 5 Classes	Openly Available [53, 55]	Supervised DL Supervised DL	Faster R-CNN Incept-ResNet-V2 YOLOv5s	mAP50 = 81.10% mAP50 = 83.40%	[52] [54]
WTB Surface Defects Detection		1,302 images 5456 x 3632 x 3	156 bounding box 12 Classes	Openly Available [59]	Supervised DL	YOLOv5s	mAP50 = 83.80%	[57]
WTB Structural Damage Detection		1,040 images pairs (RGB & IR) 6400 x 4000 x 3 640 x 512 x 1	unspecified b.box quantity 1 Class	Not Available	Supervised DL	YOLOv7	mAP50 = 98.50%	[58]

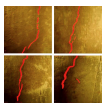
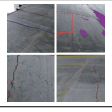
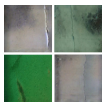
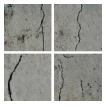
2.1.3 Hydroelectric Dam Walls

Hydroelectric dams are valuable, long-term assets that require regular maintenance to achieve their expected lifespan and maintain operational efficiency. A critical part of this maintenance is the inspection process, which serves as the foundation for identifying potential issues early. Visible light inspections, focusing on surface-level observations, are essential for detecting early signs of damage, such as cracks and discoloration. Improving inspection efficiency not only triggers timely maintenance actions but also ensures the dam operates safely, reliably, and in compliance with regulatory standards, extending its life and preventing costly repairs or failures.

Yangtao Li *et al.* [60] propose an automatic classification and localization method for structural cracks in concrete dams, based on deep neural networks and transfer learning. This approach of sliding window patch classification, also followed by ZhiLong Qi *et al.* [61], enabled pseudo-localization of crack defects on the reconstructed image, avoiding the computational effort required by object detectors or segmentation models. However, since each patch is treated independently, the relationships between patches are not considered, and the extraction of higher-level features is not possible. In their subsequent work [62], Yangtao Li *et al.*employed a lightweight semantic segmentation model for underwater crack pixel-wise identification and quantification. Kunlong Hong *et al.* [63] present a robotic solution to automatically detect and count defect instances, as well as reconstruct the surface of dam spillways, by incorporating deep learning with a visual 3D reconstruction method. Ben Huang *et al.* [64] propose an unsupervised image-to-image (I2I) translation method using the CycleGAN architecture to generate synthetic underwater images, with the goal of expanding datasets for training deep learning models. Dong Chen *et al.* [65] enhanced the texture details of underwater dam crack images using a multi-feature adversarial loss.

Table 2.4 summarizes the literature details and results on hydroelectric dam walls using RGB on Remote Operated Vehicle (ROV) platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.4: Literature Details on Hydroelectric Dam Walls employing RGB Sensors on ROV Platforms

Task Description	**Data**				**Model**			
	Sample	**Size**	**Annotations**	**Source**	**Algorithm**	**Architecture**	**Performance**	**Ref**
Concrete Dam Structural Cracks Classification		50,476 patches 200 x 200 x 3 + 10,208 patches 200 x 200 x 3	2 Classes -Crack -Background	Partially Available [66]	Supervised DL	ResNet50	F1 = 79.36%	[60]
Dam Wall Crack Segmentation		14,268 images 224 x 224 x 3 + 855 images 768 x 576 x 3	Unspecified Annotated Pixels 1 Classes	Partially Available [67]	Supervised DL	LinkNet	mIoU = 89.24%	[62]
Dam Wall Surface Defects Segmentation		450 images 848 x 480 x 4	9.7% of Pixels Annotated 6 Classes	Cloud Available [68]	Supervised DL	U-Net	mIoU = 60.3%	[63]
Dam Wall Underwater Crack Images Generation and Detection		984 images 640 x 640 x 3 + 678 images 720 x 576 x 3	Unspecified B.Box Num. 2 Classes	Not Available	Supervised DL	YOLOx	mAP = 84.33%	[64]
Dam Wall Above-Water Crack Images Generation and Detection		450 images 256 x 256 x 3	Unspecified B.Box Num. 1 Class	Available Upon Request [65]	Supervised DL	YOLOv8s	mIoU = 84.1%	[65]

2.2 Transport and Distribution Electrical Grid Assets

Electrical assets in transport and distribution systems, such as power line conductors and structures, high-voltage insulators and other components, and substations and transformers, also share a set of characteristics, like those found in power generation systems, from the automated visual inspections perspective. These assets often exhibit high variability in size, shape, texture, and position. The range of potential defects is extensive, as is the variation in the dimensions of each asset within the system, which consequently affects the optimal acquisition distance.

The IEEE 1808 standard [1] is a guide focused on the efficient collection and management of inspection and maintenance data for transmission lines assets. This guide assists utilities and contractors in developing computer-based systems for managing inspection data, ensuring accuracy, compatibility with Geographical Information Systems (GIS) systems, and integration with other enterprise software. It also emphasizes the importance of risk assessment, data analysis, and the application of AI and ML to improve the detection and management of anomalies.

The common combinations employed for data collection in Transmission and Distribution (T&D) system inspections range from manned to remote-controlled platforms, including ground-based, aerial, and wire-suspended types, equipped with sensors such as RGB cameras, Light Detection And Ranging (LiDAR), IR thermography, and Ultra-Violet (UV) imaging sensors. These combinations are used at varying capture distances and speeds, depending on the specific inspection tasks being performed. Types of inspections defined in the IEEE Std 1808 [1] include routine, emergency, comprehensive, and intrusive inspections.

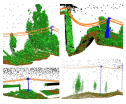
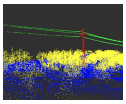
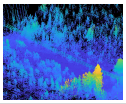
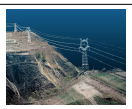
2.2.1 Power Line Conductors and Structures

Diogo Lavado *et al.* [69] introduced a 3D point cloud dataset collected using a UAV equipped with a LiDAR sensor, covering more than 40,000 km of electrical transmission systems in rural terrain. The dataset comprises 2.595 billion points and 36,000 bounding boxes across 22 classes. The five main classes (Power Line, Suport Tower, Ground, Road, and Vegetation) constitute over 88.5% of the points, with only 1.43% corresponding to electrical assets. In their latest work [70], Diogo Lavado *et al.* improved Scene-Net [71] by adding three more GENEO (Group Equivariant Non-Expansive Operators [72]) kernels and incorporating a multi-class classifier in the model output. Using a white-box feature extractor with just 540 parameters, they not only addressed the explainability problem but also set a new benchmark for parameter efficiency. Their Scene-Net V2 achieved a 50.21% mIoU with 0.26M parameters, while the state-of-the-art Point Transformer V3 [73] obtained 65.58% mIoU with 46M parameters.

Junhua Dong *et al.* [74] introduced the PSFE-Net, a semantic segmentation network for aerial LiDAR power lines inspection. They annotated 5 classes (ground, pylons, power lines, vegetation, and houses) in a subset of the AHN dataset [75], containing a transmission power line with 27.4M points. V. Valseca *et al.* [76] presented a real-time LiDAR-based semantic classification method for powerline inspection. They observed that the measured reflectivity of objects is influenced more by their geometry than by their material's inherent reflectivity properties. Alexander Kyuroson *et al.* [77] explored autonomous point cloud segmentation techniques for power line inspection, presenting a multi-step approach. Shuo Wang *et al.* [78] used a UAV to acquire 3D LiDAR data for identifying safety hazards across five transmission lines spanning 160 km in the mountainous region of Sanmenxia City, China, identifying 77 vegetation-related hazards.The LiPowerLine software, developed by GreenValley International [79], offers advanced tools for power line analysis, including automatic classification of point clouds with over 95% accuracy for transmission lines and 85% for distribution lines.

Table 2.5 summarizes the literature details and results on power line conductors and structures using LiDAR sensors on Mobile platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.5: Literature Details on PL Conductors and Structures employing LiDAR on mobile Platforms

Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
3D points Classification of Power Lines		2,595 M 3D points along 40,000 km	5 (22) Classes 36,000 B. Boxes	Code + Data Openly Available [80]	Explainable ML Supervised DL	SCENE-Net V2 Point Transformer V3 [73]	mIoU = 50.21% mIoU = 65.58%	[70] [70]
3D points Classification of Power Lines		27.4 M 3D points Unspecified Line Length	5 Classes Unspecified B. Box Num.	Annotations Not Available Data Openly Available [75]	Supervised DL Supervised DL	PSFE-Net SCD-Net [81]	mIoU = 89.90% mIoU = 88.63%	[74] [74]
3D points Classification of Power Lines		Unspecified 3D points Unspecified Line Length	4 Classes Unspecified B. Box Num.	Partially Available [82]	Unsupervised ML Supervised DL	Reflectivity Threshold + Cluster PCA PointNet++ [83]	F1 = 95.09% F1 = 83.10%	[76] [76]
Power Lines Conductors Extraction and Modelling		Unspecified 3D points Unspecified Line Length	Not Existent	Not Available	Unsupervised ML	PCA + K-D tree + DBSCAN	Not Evaluated	[77]
Power Lines Conductors Distances Assessment		Unspecified 3D points 160 km Line Length	Not Existent	Not Available	Unspecified	Unspecified	Not Evaluated	[78]

While LiDAR provides accurate spatial measurements of structures, managing 3D point clouds poses challenges due to large data volumes, complex processing, and the need for skilled interpretation. In contrast, other studies use RGB images to detect power line encroachments, focusing on simpler tasks like identifying foreign objects or bird nests, which trade some precision for easier processing.

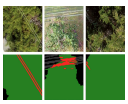
M. Gazzea *et al.* [84] propose a hybrid approach for automated monitoring of vegetation encroachment along power lines by combining high-resolution satellite imagery with LiDAR data projections as ground truth. Using satellite images for vegetation encroachment monitoring offers wide coverage, cost-effectiveness, and temporal analysis, but it faces challenges such as resolution limitations, cloud interference, processing complexity, data latency, and variability in vegetation types. M. Solis *et al.* created the VEPL dataset [85] for semantic segmentation of vegetation encroachment in power line corridors. The use of monocular images, for proximity detection tasks, imply some assumption about safety distances allowed, such as, the vertical clearance matching the Power Line (PL) height, which is not always the case. To address this limitations, other forms of data capture, besides LiDAR, might be considered. S. Rong *et al.* [86] employed advanced stereovision techniques for intelligent detection of vegetation encroachment around PL. Their proposed framework integrates binocular vision sensors mounted on PL towers, combined with embedded image processing capabilities for real-time monitoring.

Despite the limited availability of usability and impact metrics in real-world industrial scenarios, another common application of automated inspection in power line maintenance, as highlighted in the literature, is the detection of foreign objects such as bird nests, plastic bags, and balloons that become entangled in exposed conductors. Some studies [87], also include categories like fire, people, and monkeys. Zhichao Chen *et al.* [88] created a comprehensive and diverse dataset for foreign object detection on transmission lines. Recognizing the scarcity of publicly available large-scale datasets, their proposed approach combines three synthetic data sources and evaluates the contribution of each.

Several studies [89–92] have used this application scenario as a basis to explore architectural and algorithmic improvements to state-of-the-art object detectors, aiming to enhance models' ability to recognize feature patterns in image sets. However, models may become fitted (or even overfitted) to the specific data used in both training and testing, but the capacity to assess generalization depends largely on the similarities and differences between these datasets, a factor that remains challenging to quantify. Furthermore, the practical value of these applications can diminish if the primary focus shifts too heavily toward metric optimization rather than real-world utility. Achieving practical relevance requires thoughtful research into task framing to enable the proper integration of detection triggers into maintenance procedures.

Table 2.6 summarizes the literature details and results on power lines conductors and structures using RGB sensors on different platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.6: Literature Details on Power Lines employing RGB Sensors on UAV Platforms

Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Segmentation of Vegetation near Power Lines		5 Images 800 x 800 x 3 + 10 Images 400 x 400 x 3 26 km PL Length	1 Class Unspecified Pixels Num.	Not Available	Supervised ML + DL	FE + MLP + Custom CNN	F1 = 72.17%	[84]
Segmentation of Vegetation and Power Lines		532 Images 256 x 256 x 3 2.4 km PL Length	2 Classes Unspecified Pixels Num.	Openly Available [93]	Supervised DL	Unet +BCE	mIoU = 66.35%	[85]
					Supervised DL	DeepLab + Tversky	mIoU = 68.25%	[85]
Foreign Object Detection near Power Lines		14,615 images from 1920 x 1080 x 3 to 2048 x 2048 x 3	4 Classes 40,541 B. Box Num.	Code + Data Openly Available [94, 95]	Supervised DL	YOLOv8-l	mAP50 = 94.2%	[88]
					Supervised DL	YOLOv6-s	mAP50 = 93.9%	[88]
Foreign Object Detection Obstructing Utility Assets		3,140 Original Images Unspecified Resolution	9 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv5-l 1280 input	mAP50 = 86.39%	[96]
					Supervised DL	YOLOv5-x 640 input	mAP50 = 85.43%	[96]

2.2.2 High-Voltage Insulators and Other Components

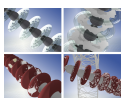
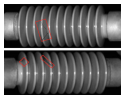
The integrity of high-voltage insulators and other components is crucial for uninterrupted service in electrical networks. UAVs equipped with RGB cameras provide a flexible solution for inspecting these assets, allowing for detailed visual assessments from various angles and heights. This technology aids in the early detection of wear and potential failures, contributing to more effective maintenance strategies.

The EPRI [20] is fostering collaboration to advance utility infrastructure inspection by promoting software development competitions and sharing labeled datasets on major data platforms, including IEEE Dataport [97]. Since the IDID-v1.2 [21] was shared in a competition context, the test set annotations were kept private. As a result, the test set contains 88 unlabeled images, while the training dataset includes 1,600 annotated images across 4 classes: String Insulator, Good Shell, Broken Shell, and Flashover Shell. However, only 400 images are unique, with the remaining images generated through offline mirroring augmentation, which may require additional care in how the validation of the trained models is performed. Jianfeng Zheng *et al.* [10] utilized the images from IDID [21] but created a new set of annotations [98] using different implicit rules. Instead of using bounding boxes enclosing the entire asset, they annotated defects present in individual insulator discs, which led to discarding the Good Disc class.

Xian Tao *et al.* [99] addressed the scarcity of defect images in real inspection environments by synthesizing defective insulator images to create the *Chinese Power Line Insulator Dataset (CPLID)* [100]. In 2018, the same authors proposed a cascade architecture [99]: first detecting the composite insulator string from the background, and then detecting the missing disc within the insulator string. Another use of the IDID [21] is in the study conducted by Bruno Souza *et al.* [101], which developed a Hybrid-YOLO model for classifying insulator defects. They employed the object detector for the insulator disc class and a classification CNN to distinguish between defective and functional assets. Zhiyong Tao *et al.* [102] presented even superior results, using the CPLID [100] and IDID [21] datasets. Reinaldo A.C. Bianchi *et al.* [103] created a synthetic high-voltage insulators dataset [104] using a game development engine and publicly available CAD models of the most common insulator types. Ebru Ergun *et al.* [105] evaluated the cleanliness of High Voltage (HV) insulators using the classification set from the synthetic dataset [104].Hericles Ferraz *et al.* [106] extended their team's previous work [103] by training and testing a deep learning model for the automated classification of electrical insulator cleanliness. Tao Zhang *et al.* [107] collected catenary insulator images captured on real railway lines to construct the Catenary Insulator Defect (CID) dataset [108]. Bojian Chen *et al.* [109] redesigned the YOLOv7 architecture to improve the detection of broken insulators in power distribution networks. Benelmostafa *et al.* [110] proposed improvements to the YOLOv8 model for detecting broken glass insulators and constructed the VPMBGI dataset [111] by merging other publicly available datasets and carefully annotating 1,010 images.

Table 2.7 summarizes the literature details and results on High Voltage (HV) Insulators using RGB sensors on UAV platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.7: Literature Details on HV Insulators employing RGB Sensors on UAV Platforms

Task Description	**Data**				**Model**			
	Sample	**Size**	**Annotations**	**Source**	**Algorithm**	**Architecture**	**Performance**	**Ref**
Functional and Defective Discs Detection in HV Insulators Strings		1688 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3	4 Classes 17,116 B. Box Num.	Open Available lewis$_i nsulator_2$021	Supervised DL	Faster R-CNN	mAP50 = 87.2%	[112]
					Supervised DL	YOLOv8s	mAP50 = 87.9%	[112]
Defective Fraction Detection in HV Insulators Discs		1600 Images from 1618 x 1080 x 3 to 7360 x 4912 x 3	3 Classes 5,373 B. Box Num.	Open Available ninja$_d ataset$	Supervised DL	YOLOv7	mAP50 = 90.1%	[10]
					Supervised DL	Improved YOLOv7	mAP50 = 93.8%	[10]
HV Insulators Strings and Missing Discs Detection		848 Images 1152 x 864 x 3	2 Classes 1,325 B. Box Num.	Open Available cplid$_d ataset$	Supervised DL	YOLOv8n	mAP50 = 99.4%	[102]
					Supervised DL	SnakeNet	mAP50 = 99.5%	[102]
HV Insulators Dirty Defect Classification		47,286 Images 512 x 512 x 3 + 14,424 Images 512 x 512 x 3	4 Classes of Dirty HV Insulators	Open Available [113]	Supervised DL	Custom CNN	mIoU = 97.32%	[106]
					Supervised DL	VGG16 + RF	mAP50 = 98.54%	[105]
Catenary Insulators Defective Pixels Classification		4,293 Images 512 x 512 x 1	7 Classes 10% Defects	Data + Code Open Available [108]	"Unsupervised" DL	FastFlow [114]	F1 = 63.07%	[107]
					"Unsupervised" DL	MAE [115]	F1 = 68.89%	[107]
MV Insulators Broken Defect Detection		1,236 Images 512 x 512 x 3	1 Class 2,438 B. Box Num.	Open Available [116]	Supervised DL	YOLOv7	mAP50 = 78.5%	[109]
					Supervised DL	ID-YOLOv7	mAP50 = 85.7%	[109]
Object Detection HV Insulators		1100 Images 640 x 640 x 3	1 Class Unspecified B. Box Num.	Data + Code Open Available [111, 117]	Supervised DL	YOLOv8m	mAP50 = 92.0%	[110]
					Supervised DL	YOLOv8-Gold	mAP50 = 94.1%	[110]

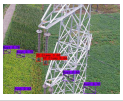
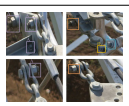
Acknowledging the lack of publicly available real-world datasets for power line components,

André Silva *et al.* [118] introduced the Power Line Asset Inspection Dataset, InsPLAD [119]. The same authors, in [120], explored attention modules for image-level anomaly detection using, among other publicly available datasets, a subset of their own dataset [119], containing 11,662 images, of which 402 were defective samples used only for testing. Chuanyang Liu *et al.* [121] argue that rust defects on power components pose significant hidden dangers to the safe operation of transmission lines. To address this, they modified the YOLOX model architecture to detect rust on dampers and clamps. Jun Wu *et al.* [122] modified a YOLOv5 model to detect corrosion in transmission line components, integrating several architectural components (WA-PANet, PSA, RFENet, and EIoU Loss). Frames were extracted from a single video, resulting in 2,705 images with a resolution of 5184×3888. Anicetus Odo *et al.* [123] investigated the effects of occlusion, scale, and multi-task detection in asset management of PL. The original dataset, provided by Northern Powergrid, derives from helicopter inspections of three power lines, consisting of 5616×3744 images from eight different views of 7,219 PL tower assets.

Motivated by the challenge of detecting ultrasmall defects in bolts during transmission line inspections, Peng Luo *et al.* [124] developed a method to fully leverage the high-resolution images commonly captured in modern inspections. Despite requiring longer inference times, their Ultrasmall Bolt Defect Detection Model (UBDDM) outperformed other state-of-the-art (SOTA) architectures in precision and recall, achieving a 78.6% mAP compared to 47.2% with Faster R-CNN. Yuting Wu *et al.* [125] studied another application of automated inspections for electrical grid assets by training a modified YOLOv8 model to detect damages in PL conductor cable wiring.

Table 2.8 summarizes the literature details and results on several power lines components using RGB sensors on UAV platforms, with a focus on crucial details for deep learning model comparison, such as task definition and dataset characteristics.

Table 2.8: Literature Details on Power Lines Components employing RGB Sensors on different Platforms

Task Description	**Data**				**Model**			
	Sample	**Size**	**Annotations**	**Source**	**Algorithm**	**Architecture**	**Performance**	**Ref**
Defective Classification of PL Components		10,607 Images 1920 x 1080 x 3 39,150 patches	17 Classes 28,933 B. Box Num. 5 Classes w/ Defects	Open Available [119]	"Unsupervised" DL	DifferNet	AUC = 90.5%	[118]
					"Unsupervised" DL	AttentDifferNet	AUC = 94.34%	[120]
Rust Detection on PL Dampers and Clamps		3400 Images 416 x 416 x 3	4 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv7	mAP = 91.65%	[121]
					Supervised DL	RD-YOLO	mAP = 92.33%	[121]
Rust Detection on PL Dampers and Fittings		2705 Images 5184 x 3888 x 3	4 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv5	mAP = 90.15%	[122]
					Supervised DL	PWR-YOLOv5	mAP = 95.37%	[122]
PL Tower Health Condition Classification		Unspecified Images Num. 5616 x 3744 x 3 386,401 patches Unspecified Resol.	2 Classes Healthy Unhealthy	Not Available	Supervised DL	ResNet50v2	F1 = 84.13%	[123]
					Supervised DL	EfficientNetB0	F1 = 86.51%	[123]
Bolt Defect Detection on PL Fittings		1,852 Images Unspecified Resol.	6 Classes 14,635 B. Box Num.	Not Available	Supervised DL	Faster R-CNN	mAP = 47.2%%	[124]
					Supervised DL	UBDDM	mAP = 78.6%	[124]
PL Conductors Damage Detection		1,318 Images 1080 x 1080 x 3	2 Classes 1,512 B. Box Num.	Open Available [126]	Supervised DL	EfficientDet	mAP50 = 90.1%	[125]
					Supervised DL	Modified YOLOv8	mAP50 = 91.1%	[125]

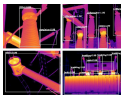
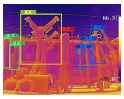
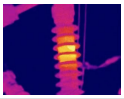
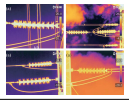
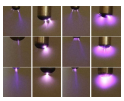
2.2.3 Substations and Transformers

Inspections of substations and transformers require reliable monitoring systems due to their critical role in T&D. Early detection of sudden temperature rises, which often precede equipment failures, is essential to prevent assets damage and power outages. Fixed platforms equipped with IRT and UV sensors enable real-time analysis of these assets, facilitating the immediate identification of anomalies such as overheating or insulation degradation. This proactive approach to monitoring significantly reduces downtime and enhances operational efficiency.

Sheng Han *et al.* [127] developed an application for diagnosing electrical equipment using a smart thermographic camera. Jianhua Ou *et al.* [128] used an improved Faster R-CNN model to detect substation electrical equipment in infrared images. Bin Wang *et al.* [129] propose an automatic diagnosis method using instance segmentation and temperature analysis of infrared insulator images. Recognizing the lack of a publicly available dataset of infrared images of insulators, Jinhui Zhou *et al.* [130] used a FLIR T640 infrared camera to capture 2,540 images with a resolution of 640x480, containing insulators on 110 kV transmission lines.

Besides disturbing the environment, wildlife, and nearby communications, the corona effect, caused by the ionization of the air, results in power loss and damage to assets. Prolonged exposure leads to wear and tear on insulating equipment through material degradation, ultimately causing energy inefficiencies and equipment failures. Halil Ibrahim Uckol and Suat Ilhan [131] studied the visual light corona discharge patterns from different electrode tips under various voltage levels. To overcome the limitations of detecting visible corona discharge during daylight, Noushin Davari *et al.* [132] proposed an inspection system for identifying UV-invisible defects in (Medium Voltage (MV)) PL.

Table 2.9: Literature Details on Substations and Transformers employing different Sensors and Platforms

Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Substations Assets Health Condition Diagnostic		12,683 Images 640 x 480 x 1	16 Classes 18,146 B. Box Num.	Not Available	Supervised DL	Retina-Net	mAP50 = 85.3%	[127]
					Supervised DL	Modified YOLOv4	mAP50 = 86.4%	[127]
Substations Assets Detection		1,263 Images 600 x 600 x 1	5 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv5 CSPdarknet53	mAP50 = 91.85%	[128]
					Supervised DL	Faster R-CNN VGG16	mAP50 = 92.78%	[128]
HV Insulators Temperature Measurement		1,200 Images 1024 x 1024 x 1	1 Classes Unspecified B. Box Num.	Not Available	Supervised DL	Mask R-CNN ResNet101-FPN	mAP50 = 77%	[129]
HV Insulators Segmentation		2,540 Images 640 x 480 x 1	1 Classes Unspecified B. Box Num.	Not Available	Supervised DL	BSIS ResNet101-FPN	mAP50 = 95.19%	[130]
Corona Detection and Discharge Mode Classification		741 Images 500 x 500 x 3	4 Classes Unspecified B. Box Num.	Open Available [133]	Supervised DL	YOLOv8m	mAP50 = 99.3%	[134]
					Supervised DL	ShuffleNet	F1 = 100%	[134]
Corona Detection and Power Equipment Classification		462 Images 4 Channels Unspecified Resolution	7 Classes 4,486 B. Box Num.	Not Available	Supervised DL	Faster R-CNN	mAP = 78%	[]
					Supervised DL	Optimized Faster R-CNN	mAP = 82%	[]
Substations Assets Instance Segmentation		1,660 Images 640 x 480 x 3 4672 x 3504 x 3	15 Classes 50,705 B. Box Num.	Data + Code Available [135, 136]	Weak-Supervised DL	DeepLabV3+ GrabCut SDI	IoU = 62.9%	[137]

2.3 Summary and Insights

Visual inspection is a cornerstone of non-destructive testing (NDT) in electrical asset management. Traditionally reliant on human observation and passive electromagnetic signals, visual inspections assess the condition and integrity of components such as cables, connectors, switches, and circuit breakers. This approach enables early detection of physical damage, wear, improper installation, and environmental degradation—ensuring compliance with safety standards and operational efficiency across industrial and commercial settings.

Recent technological advances have revolutionized these inspections by integrating digital data acquisition techniques. Modern platforms—ranging from high-resolution cameras and laser scanners to UAVs (unmanned aerial vehicles)—capture precise, multi-modal data (including visible, thermal, and depth information) at various scales. This systematic approach not only increases the efficiency of data collection but also enables long-term storage and comparative analysis. The shift from traditional manned inspections (e.g., helicopter surveys) to UAV-based systems enhances safety, scalability, and consistency by minimizing human subjectivity and risk.

However, while data acquisition has become more automated, data processing remains a significant challenge. The large volume of heterogeneous data necessitates advanced computer vision (CV) techniques and deep learning (DL) models to accurately detect anomalies. Early implementations using convolutional neural networks (CNNs) demonstrated the potential to outperform traditional CV methods, especially in detecting subtle defects. Yet, these generic models, often trained on datasets like ImageNet, fall short when applied to domain-specific tasks in electrical inspections. The complexity of electrical systems and the unique characteristics of asset defects require customized, domain-specific models that leverage high-quality, meticulously annotated data.

To address these challenges, a dual approach has emerged. The model-centric approach focuses on optimizing neural network architectures and learning algorithms to improve performance. In parallel, the data-centric approach emphasizes the systematic engineering of data—enhancing annotation detail, consistency, and granularity. By incorporating precise spatial labeling (using various annotation scales, from large bounding boxes to detailed polygons) and refining categorical definitions, this strategy improves model accuracy even with smaller datasets. Such data engineering is critical for guiding neural networks toward more robust representations of complex defect patterns.

The convergence of these strategies supports the development of automated inspection workflows that are not only faster and more cost-effective but also more reliable. The integration of computer vision, AI-driven anomaly detection, and predictive maintenance algorithms enables proactive decision-making and helps prioritize maintenance actions. This transformation of traditional inspection methodologies into data-driven, automated processes is pivotal for ensuring the safety, reliability, and longevity of electrical grid assets.

In summary, by leveraging advances in digital data acquisition, UAV technology, computer vision, and both model-centric and data-centric AI approaches, this work lays the foundation for next-generation automated visual inspection systems. These systems promise to significantly enhance grid maintenance operations, reduce costs, and improve overall system performance in an increasingly complex and demanding environment.

Chapter 3

Proposed Work Plan

This chapter provides a detailed description of the proposed research plan, including the specific tasks, objectives, and methodology that will guide the project. It outlines the scope of the work and explains how the research will be conducted, with a focus on practical implementation and collaboration with industry partners. The proposed work plan aims to provide a roadmap for achieving the research objectives and contributing to the field.

3.1 Overview

This research focuses on the development and validation of deep learning models for automated visual inspection of electrical grid assets. Conducted in collaboration with EDP Labelec, the project aims to integrate these models into the operational workflow of a Distributed System Operator (DSO), ensuring practical applicability.

The proposed approach involves developing efficient techniques for automated visual inspection of electrical grid assets, creating new Computer Vision (CV) deep learning models tailored for this application. The quality of the data, with diverse image modalities, will be guaranteed by human validation and an active learning approach using the expertise of a Distributed System Operator (DSO).

To further improve the inspection process, and envisaging real-time operation and edge processing, data collection will be validated at capture time by a CV embedded system, using low complexity, neuromorphic-oriented models.

The developed models will be validated in the real-world conditions through the integration into the DSO processes, aiming to evaluate their effectiveness and reliability.

3.1.1 Graphical Abstract

Figure 2 illustrates the current workflow of electrical grid inspections, encompassing the full spectrum of activities, from high-level asset management strategies and inspection planning to asset quality reporting and maintenance operations. This figure highlights key stages, particularly on-site data collection and analytical data processing, which currently represent bottlenecks in the inspection workflow. While increasing data volume can improve assessment quality, it also introduces redundancy, reducing efficiency when processed through labor-intensive methods.

The proposed system introduces an integrated ecosystem that enhances data collection by validating its quality and quantity in real time through an embedded vision system. Unlabeled data is processed within a multimodal framework that incorporates complementary information about the electrical system under inspection, such as time-series and spatial data. The analytical pipeline follows an active learning approach, continuously refining its models as the dataset expands. Human validation is incorporated based on decision uncertainty, ensuring that expert review is prioritized where needed. This approach optimizes efficiency by reducing manual effort while maintaining accountability. Additionally, the system enables seamless reporting of collected data through an asynchronous digital twin platform, facilitating informed decision-making and improving the overall inspection workflow.

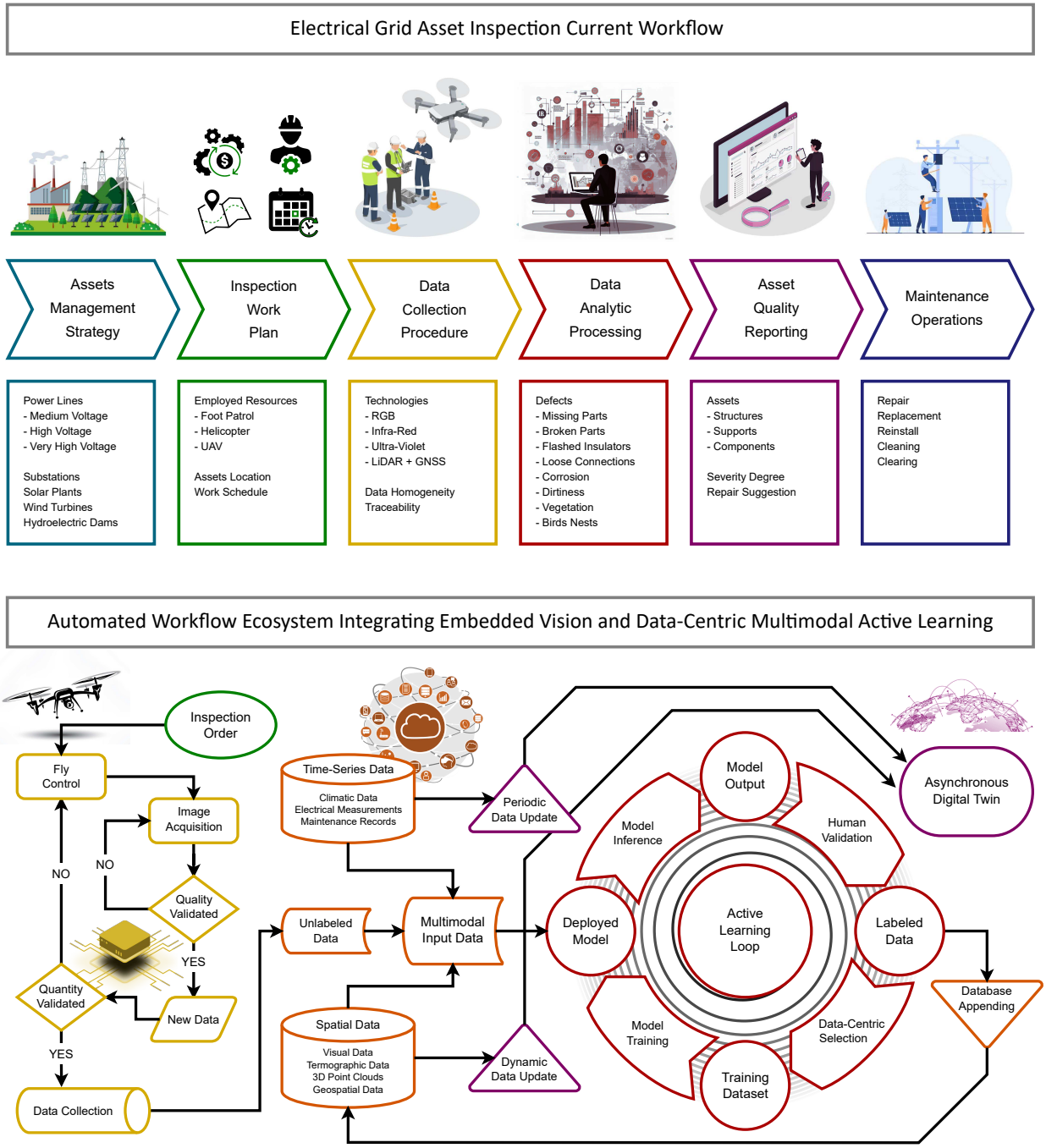


Figure 2: Overview of the current electrical grid inspection workflow and the proposed automated ecosystem integrating embedded vision and data-centric multimodal active learning technologies.

3.2 Detailed Description

This project, to be developed in a non-academic environment, aims to develop cutting-edge computer vision technologies based on advanced deep learning models, that can enhance the automatic inspection process of electrical grid assets. Deep learning models have demonstrated remarkable progress, driven by advancements in computational power, complex architectures, and data availability. However, existing public datasets lack the quantity and quality of data needed for reliable model creation, especially in the electrical grid assets' domain. The most effective solution involves collaborating with a Distribution System Operator (DSO) such as EDP, to gather a very large, high-quality dataset.

EDP Labelec has several teams conducting virtual inspections using UAVs equipped with high-resolution RGB, IR, and LiDAR sensors, which have collected vast amounts of data over the last few years. Labelec employs specially trained personnel who assess the operational conditions of grid assets through analysis of their digital representations, which are crucial for creating dataset annotations. The company possesses the flexibility and autonomy to adopt innovative methodologies, both in image processing and in the acquisition and collection process, providing the infrastructure to test the findings and developments outlined in the work plan under real-world conditions.

Instituto de Telecomunicações, in particular the MSP-CO group, has up-to-date facilities to support development and testing of Machine Learning algorithms as described in the work plan. The labs have been supported by NVIDIA as an NVIDIA Research Center and are equipped with several high-performing workstations with the latest generation GPU hardware. The laboratories have several large capacity data storage servers, currently hosting datasets for internal and JPEG standards committee use. These data servers will be crucial to store the very large datasets of RGB and IR Images, as well LiDAR point clouds to be captured and processed as described in the work plan.

The collaboration between academia and industry yields mutual benefits and can contribute to the overall success of this research project. The advantages of the participation of a specialized non-academic partner are three-fold. First, the non-academic partner can be an expert source to contribute to the definition of the global objectives and directions of the research to be carried out, ensuring high industry relevance of the scientific research. Second, academic institutions gain access to a large quantity of real world data, allowing to create a vast electrical grid asset dataset, vital for training and validating advanced computer vision models. The non-academic partner can handle data quantity and expert human classification, while the academic side ensures technical data quality and annotation accuracy. Through rigorous validation processes, academia ensures the quality of the data, including image and instance annotations, enhancing the effectiveness of the models. Third, this collaboration enables the validation of models in real-life testing environments, testing field deployment conditions and ensuring robustness and generalization.

The optimization of inspection processes using advanced computer vision technologies results in significant cost reductions and increased efficiency for the industry. Ultimately, this partnership aims to enhance the reliability of the electrical grid by detecting and addressing issues more effectively, minimizing downtime, and ensuring a stable infrastructure for consumers and businesses alike. Thus, this collaborative effort drives advancements in computer vision technology and tangible improvements in the electrical energy industry, through the improvement of grid efficiency and reliability.

3.2.1 Objectives

The main objective of this research is to develop new efficient computer vision techniques based on innovative deep learning models and active learning to automate the visual inspection processes of electrical grid assets. To achieve this objective, the research will focus on solving current problems that are holding back advancements in this specific area. Domain-specific algorithms and model architectures especially designed for the semantic hierarchic structure that characterize the electrical grid assets will be investigated. High quality dedicated datasets will be created, with expert annotations reflecting the relevant features that support current direct visual inspections.

Another objective is to explore the joint use of different image modalities and other signals to improve the performance of fault detection. Time-series data, such as electrical measurements, in addition to spatial information like thermographic patterns, might improve the detection of loose connections, overloaded circuits, or faulty equipment. Furthermore, temporal and spatial data, such as geolocation and point-clouds, can be used to create accurate 3D representations in a Digital Twin for improved remote inspections and maintenance management. An additional parallel objective is to create low-complexity, low-resource models possibly based on neuromorphic tenets, that can be integrated in UAVs or other embedded platforms for edge processing, enhanced by the use of event cameras to reduce latency and optimize data collection.

It is expected that the integration of the research results into the grid monitoring process will enable a transition from a human based workflow to a highly automated monitoring solution. Other objectives of this research include the validation of the developed models in a real-world environment, the sharing of high quality datasets and the dissemination of the knowledge acquired. This dissemination aims to optimize processes within the industry, improving efficiency and reducing operational costs, while contributing to the overall reliability of the electrical grid infrastructure.

3.2.2 Project Main Tasks

To comprehensively address the outlined objectives, this proposed research will explore several key directions. A timeline distribution of the tasks is provided in Figure 3, detailed by the description of the following tasks to be performed at IT and Labelec.

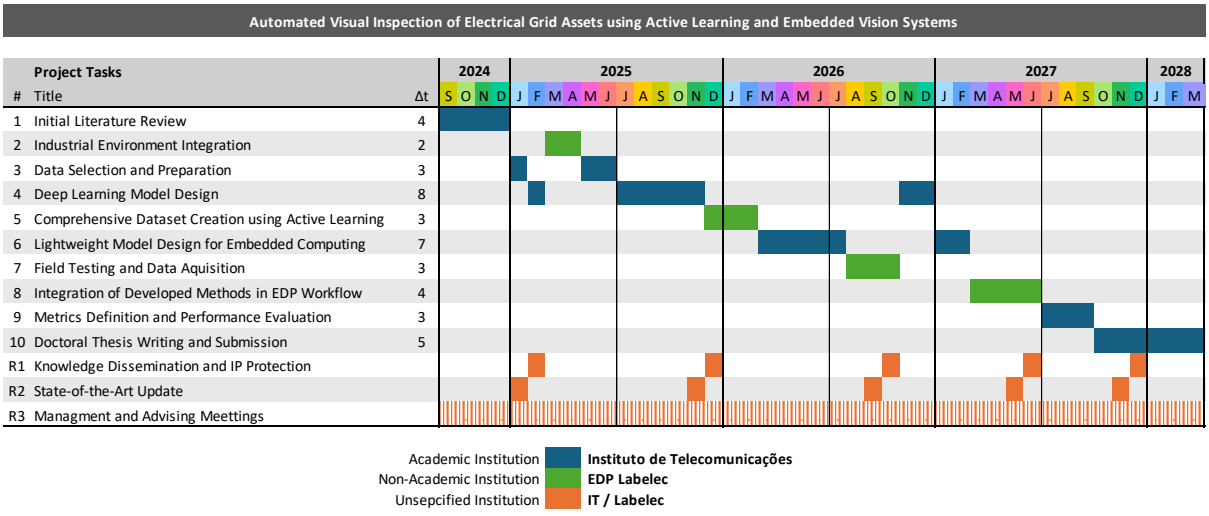


Figure 3: Chronogram illustrating the timeline and overview of project tasks

1. **Initial Literature Review (2 months)** – Review the latest literature to identify research gaps and emerging trends in deep vision techniques and procedures, and in electrical grid asset inspection technologies and processes. This task was already completed and has been partially included in Section 2.1 and Section 2.2 of this document, while an associated journal publication is being prepared for submission.

2. **Industrial Environment Integration (2 months)** – Onboarding at EDP Labelec to become familiar with operations and tools to better understand important details of the inspection process and data analysis. Additionally, it aims to clarify and investigate the full potential regarding the value of combining visual data modalities and power transition data in a multimodal learning approach. Contacts with EDP Labelec already confirmed the start of this task in the proposed timeslot.

3. **Data Selection and Preparation (2 months)** – Update of datasets using a data-centric approach for training and testing deep learning models, which includes defining object bounding boxes, segmentation masks, and categories for relevant assets in the electrical grid inspection domain. During this stage, experiments will be conducted to evaluate the quality of the annotations, exploring the potential benefits of implementing explicit hierarchical nested semantic structure categories when dealing with limited data diversity, and assessing the potential advantages in fine-grained information of pixel classification in instance segmentation contrasting with the widely adopted window-shaped object detection method, especially in cases of overlapping irregularly shaped objects. Valuable insights for this task have been obtained in the exploratory work described in Section 4.2 and Appendix A.

4. **Deep Learning Model Design (12 months)** – Development of multimodal defect detection methods tailored to the specific requirements of electrical grid asset inspection, striving to achieve or surpass current state-of-the-art methods with equivalent or reduced computational complexity. While collecting high-resolution image and video is crucial for capturing important details, its processing is computationally expensive and inefficient due to the large number of irrelevant pixels. Off-the-shelf solutions often down-sample images for faster processing, sacrificing feature extraction. To address this, alternate processing models will be developed, that discard redundant information by cropping images around the assets of interest. This sequential cropping process follows a hierarchical nested structure of predefined classes, allowing for more efficient feature extraction. Additionally, it will be considered the use of super-resolution methods [138] to further enhance the extraction of small features.

5. **Comprehensive Dataset Creation with Active Learning (3 months)** – Comprehend the annotation of unlabeled data optimized by an active learning strategy, which involves using a trained model to infer annotations for new data. By subjecting the results to subsequent human validation, it allows the corrected mispredictions to promptly integrate into the training dataset, reducing the labor needed to create a large annotated dataset. Privacy concerns limit shared data in quality control industries. Robust security measures are essential to safeguard sensitive information and enable collaboration [139]. The project aims to automate the filtering, blurring, or pixelating of sensitive information such as faces, license plates, or landmarks, using advanced computer vision methods. This post-processing ensures the datasets follow GDPR mandates and can be used for research, development and grid monitoring.

6. **Lightweight Model Design for Embedded Computing (12 months)** – To propose algorithmic and architectural improvements to optimize the balance between model complexity and required computational resources of the developed deep learning models to

enable real-time execution and edge processing on resource-constrained platforms. This efficient model version will be used in a prototype to further optimize the remote inspection process, aiming to address not only the processing of images but also their acquisition. Low-complexity methods based on neuromorphic processors will be evaluated to allow the use of event camera [140] outputs, in asset defect detection and classification.

7. **Field Testing and Data Acquisition Validation (3 months)** – Assess the effectiveness of the prototyped methods developed in prior phases through lab simulations and field conditions. Confirming the capability of the embedded vision system to timely validate the quality of images by assessing their visibility, focus, and clarity, and to promptly verify the quantity of images by ensuring sufficient coverage of each object's surface captured over the collected images.

8. **Integration of Developed Methods into the EDP Workflow (4 months)** – Deployment of the validated model, ensuring compatibility with existing systems and processes, to test its performance in a real-world scenario. Additionally, this phase aims to provide guidelines for developing a user-friendly GUI. This interface will facilitate the inspection process by providing intuitive access to key functionalities such as image uploading, annotation tools, and inspection result visualization.

9. **Metrics Definition and Performance Evaluation (2 months)** – Comprehensive evaluations to quantify optimization impacts on KPIs such as cost reduction and inspection efficiency. This will also include defining customized evaluation metrics tailored to asset management. To effectively inspect the quality of assets, multiple photoshoots are taken from various angles. Identifying defects in just one photo or all photos of the same asset, has the same practical implications, triggering maintenance procedures such as equipment replacements. However, if the count of defects does not consider the multiplicity of detections, the number of assets planned for replacement would be incorrect. Process performance metrics can be used to fine tune and improve the developed models.

10. **Doctoral Thesis Writing and Submission (6 months)** – Represents the culmination of the research journey, synthesizing research findings, insights, and contributions into an organized structure. This includes the interpretation of data, critical analysis, development of compelling arguments, and engagement in iterative cycles of revision and refinement to create a comprehensive document. The final stage involves undertaking an oral examination to defend the research conducted and the conclusions drawn.

Recurring Tasks:

1. **Knowledge Dissemination and IP Protection** – Dissemination of research findings through the submission of journal articles, conference presentations, and patent applications, protecting the intellectual property generated during the project.

2. **State-of-the-Art Update** – Conducting a regular reassessment of the latest literature to identify relevant works and emerging trends in deep vision techniques and procedures, and in electrical grid asset inspection technologies and processes.

3. **Management and Advising Meetings** – Regular meetings with the advisors to discuss progress, receive feedback, and address challenges encountered during the research. These meetings will include local weekly sessions, as well as monthly online meetings with academic and non-academic advisors to ensure effective communication and collaboration throughout the project.

Chapter 4

Developed Work

This chapter summarizes previous and current research conducted in the field of automated visual inspection, focusing on deep learning architectures and annotation procedures. The chapter aims to demonstrate the progress made so far and the experiences gained, which will inform the future work outlined in the following sections.

In previous research, several deep learning architectures were implemented for grid asset inspection tasks, including CNN-based models for defect detection. These models were developed using popular frameworks like TensorFlow and PyTorch, with an emphasis on ensuring scalability and robustness. Early results showed that deep learning approaches could outperform traditional computer vision methods, particularly in detecting subtle defects and anomalies.

These architectures served as the foundation for the current research, where the focus has shifted to improving model efficiency and incorporating domain-specific features such as semantic hierarchies for asset classification. This transition was motivated by findings from the MSc research, which identified key challenges in model generalization, real-time inference constraints, and dataset biases. Addressing these challenges has become a central objective of the PhD.

Another critical aspect of the work involved evaluating the impact of different annotation procedures on model performance. Early experiments demonstrated that the granularity of annotation, especially when employing nested semantic structures, could improve defect detection accuracy in instances with overlapping or irregularly shaped objects. This aligns with the MSc findings, where annotation inconsistencies were observed to significantly affect model reliability, reinforcing the need for expert-driven annotation strategies.

By building upon these insights, the current research refines model architectures and annotation methodologies to enhance both accuracy and deployment feasibility. The lessons learned from the MSc research have played a pivotal role in shaping the objectives of the PhD, particularly in optimizing deep learning models for real-world inspection scenarios.

4.1 Implementation of Standard Architectures

This research evaluated the performance of computer vision based on deep learning models for automated visual inspection of electrical grid assets, validated with real-world data. The developed work presents an example application of automated visual inspection of HV insulators, detecting defects in disc insulators in visible light images, using two state-of-the-art deep learning models. YOLOv8s achieved a mAP@50 of 87.9%, while Faster R-CNN X101-FPN achieved 87.2% for the same metric. The findings highlight the advantages and limitations of automated visual inspection, enabling utility companies to benefit from higher efficiency inspection processes, reducing the costs and improving the reliability of electrical grid maintenance. The work described here was carried out during the MSc, where the challenges and limitations encountered provided the foundation for the PhD research, shaping its central theme and proposal.

4.1.1 Methodology

This work focuses on the core of an automated inspection pipeline, receiving high resolution RGB images, collected by UAVs in real life scenarios of electric power T&D, containing different types and amounts of HV insulators, and generating a localization bounding box with a respective class of the predicted defect. The assets addressed in this study are the ceramic discs of each *Cap and Pin* insulator. Apart from the default background class, the implemented model classifies each insulator as either functional (good), defective disc (broken), or defective shell (flashover).

To conduct our experiments, we used an organized Insulator Defect Image Dataset (IDID), released by the EPRI [141] and also published on *IEEE DataPort* [142], that comprises 1600 high-resolution images, accommodating 17116 annotated instances (78% good, 15% flashover and 7% broken). Our observations suggest that the 1600 annotated image files, contain 400 unique images along with 3 mirrored copies. This information led us to avoid performing a random validation dataset split as has been done for example in [143], and not explicit in other cases [144] [145]. Creating a validation directory, carefully selecting 15% of the annotated images, ensured that no mirrored copies of the training dataset were included, ensuring the validation results integrity. Upon closer examination of some images and their annotations, such as the one presented in Figure 4, it is noteworthy that, disc insulators cropped by the edge of the image, and *good* insulator discs, adjacent to *broken* ones, are not annotated. This decision potentially arose from the overlapping context of the bounding boxes.



Figure 4: Example of missing annotations in the annotated IDID

4.1.2 Results and Discussion

The presented results were automatically obtained by inferring the validation dataset split, as it contains annotations. A direct comparison reveals that YOLOv8 has surpassed Faster R-CNN in the most demanding metric (mAP50-95) across all classes. However, it's not always the case. A few examples exist, where one outperforms the other. To compare the validation results obtained in this work with those obtained in other studies that used the same dataset, a more common metric was also used, the mAP50, as other metrics were not available in the assessed literature. Table 4.1 summarizes the validation results of the Mean Average Precision at the 0.50 IoU threshold level in a class-wise comparison.

Table 4.1: Class-wise validation mAP@0.50 IoU comparison

mAP	**Model**	**Class**			
		All	Good	Broken	Flashover
@0.50	YOLOv8s	**0.879**	0.929	0.827	**0.880**
	Faster R-CNN	0.872	**0.943**	**0.830**	0.842
	ML YOLOv5	0.776	0.834	0.748	0.745
	YOLOv5	0.327	0.760	0.000	0.220

This summary highlights the marginal superiority of the Faster R-CNN model in the *good* and *broken* classes, indicating that, while YOLO may achieve better bounding box regression accuracy, Faster may exhibit higher classification accuracy in some cases. Additionally, the results of the models implemented in this study outperform others found in the literature, more precisely, the results obtained by Das *et al.* [144], using YOLOv5, and the results obtained using a new variant named ML YOLOv5, proposed by Wang *et al.* [145].

By observing the inferred images in the testing dataset, which share similar characteristics to the annotated dataset, it is clear that both of the models (YOLOv8s in Figure 5, successfully predicted the considered instances, identifying both defective insulator disks and functional ones in the majority of the images.

Another non-annotated dataset for MV power lines, owned by EDP LabElec, was used for testing the model capabilities. Some images of this dataset were subjected to inference by both models, validating the potential of the approach conducted in this work in detecting and classifying disc insulator objects. Cropped images resulting from YOLO inference on the EDP LabElec dataset are presented in Figure 6.

This inference test on a dataset with different characteristics from the one used for training revealed that the trained YOLO model achieved better generalization capabilities. The Faster model struggled to make valuable detections on images with blurred objects, primarily because the training data consists of close planes with the focal point on the object's surface. One future development is to perform enough expert annotations to create a large-scale training set that will enable to improve the model through a fine-tuning process.

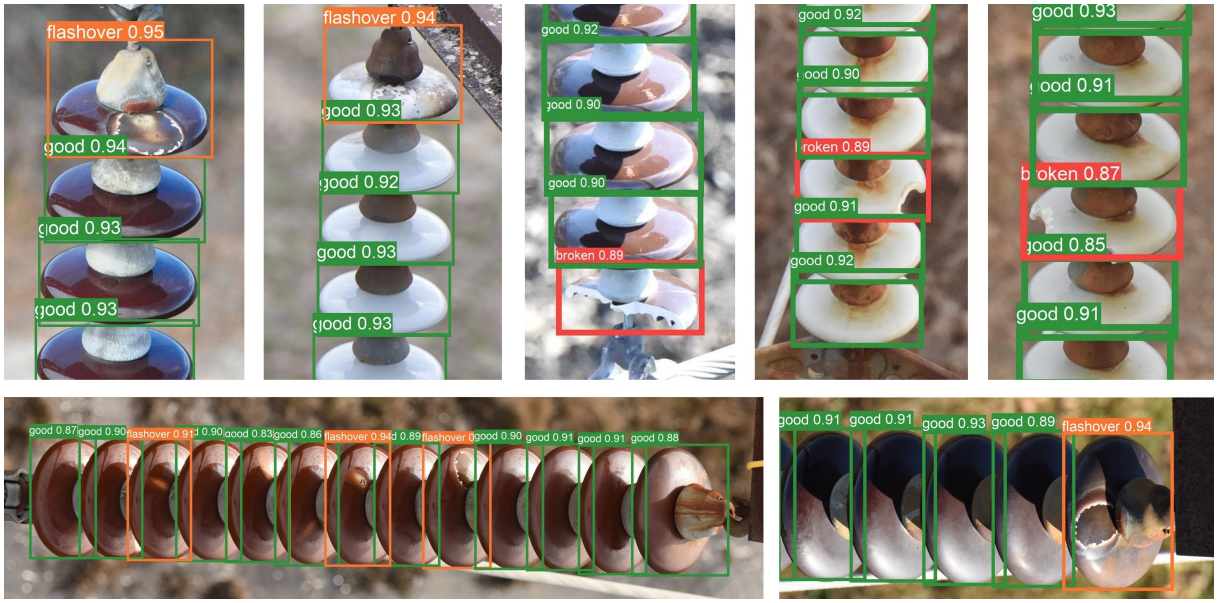


Figure 5: Correct predictions on inference test using YOLOv8s



Figure 6: Example inference test for EDP LabElec data using YOLOv8s

4.1.3 Conclusion

This work evaluated the potential of deep vision models to be used as a component for automated visual inspection of electrical grid assets, focusing on HV insulators. This performance analysis is conducted as a first step for the development of low complexity models targeting the integration in smart grid operations, aiming to improve the maintenance efficiency of electrical T&D systems. This inspection concept used for HV insulators can be expanded to other grid assets. Defects in HV insulators were detected using SOTA deep learning models to process visible light images sourced from a publicly available dataset, that contains 1600 images with over 17000 annotations of insulator discs. Both models presented good performance results, for a fully automated approach, with a mean average precision (mAP@50), for the three considered classes (*good* insulator; *broken* disc; and *flashover* shell), of 87,9% and 87,2% on the YOLOv8s and Faster R-CNN X101-FPN, respectively.

4.2 Annotation Procedure Impact Evaluation

This study analyzes the impact of annotation detail and quality on model performance, fine-tuning several YOLOv11 models, in a data-centric approach of improving the quality of a high-resolution HV insulator image set, that is evaluated based on three vectors: consistency of annotations (mitigating noise with strict labeling); spatial granularity (large box, small box, and polygon); and categorical definition (varying the number of annotated classes). The experimental results evaluate performance across different data quality levels, demonstrating improved test accuracy through enhanced label precision. Furthermore, the study concluded that higher data quality reduces the need for large quantities of annotated data. The conclusions of this work are important to define updated guidelines for optimal labeling effort management. For further details, refer to Appendix A. This research, undertaken in a scholarship between the MSc and PhD, provided key foundations for the doctoral study.

4.2.1 Methodology

This work analyses the impact of annotation quality and detail level on model performance, evaluating how varying degrees of annotation precision affect the accuracy and effectiveness of a State-Of-The-Art (SOTA) Deep Vision model. Rather than validating architectural or algorithmic changes to the model, the study uses a data-centric approach, iteratively improving the data quality used to fine-tune several pre-trained variants (small, medium and large) of the 11th YOLO version [146]. The experiments in this study were carefully designed to evaluate the impact of various annotation strategies. This resulted in a comprehensive set of experiments, that combine various ultra-hyperparameters, ranging from labeling strategy and cross-validation dataset splitting to model complexity. To account for the inherent randomness in the model fitting process, multiple experiments were conducted. Cross-validation was employed to assess the influence of image selection in the training split, while the effect of model complexity was analyzed using different variants of SOTA models.

Annotation Consistency experiments evaluate the impact of varying the level of Quality Control (QC), ranging from *basic* to *careful*. The *basic* QC policy level represents labeling only obvious defects, while the *careful* QC policy level includes challenging or subtle defects. The *reference* [98] was used as a central threshold for the QC level. Table 4.2 provides examples of annotation quality levels and instance counts distribution.

Spatial Granularity influence is assessed in the object shape dimension with a domain ranging from box_L to *polygon*. The box_L level represents large bounding boxes encompassing the entire defective disc. At the other extreme, the *polygon* level tightly fits the object shape. The small box box_S was obtained by inscribing the respective polygons. Table 4.3 presents a visual sample and instance counts for each level.

Categorical Definition assessment was conducted in the classes discarded dimension, with a domain ranging from *none* to *good*, which includes the 4 original classes (string, good, broken, flash) [21], and the 2 defective classes respectively. An intermediate level includes 3 classes (good, broken, flash), excluding only *string* insulators. Each variation is illustrated in Table 4.4, which also provides instance distributions plots.

Each model training session was conducted over 250 epochs, with images resized with padding to input size of 1280×1280 and automatic batch size. The first two layer blocks were frozen, and the default augmentation parameters were applied.

Table 4.2: Methodology description of Annotations Consistency

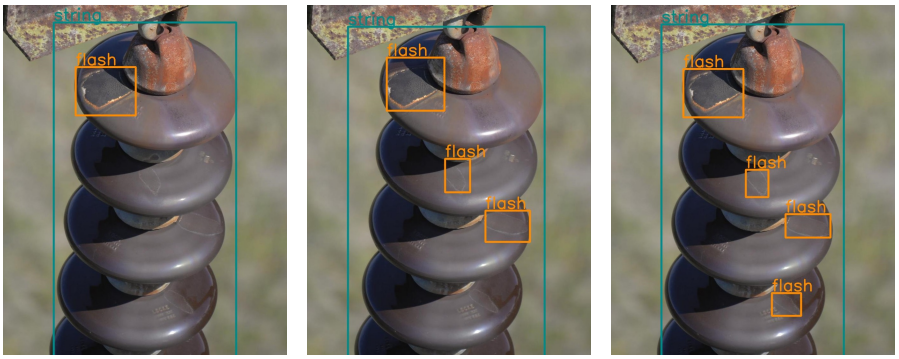
Quality Control	**Basic**	**Reference [98]**	**Careful**
Visual Sample			

Table 4.3: Methodology description of Spatial Granularity

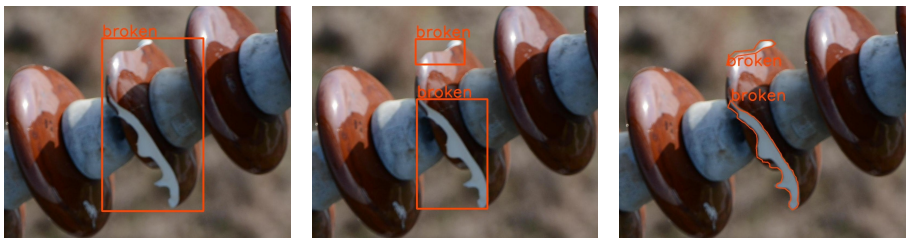
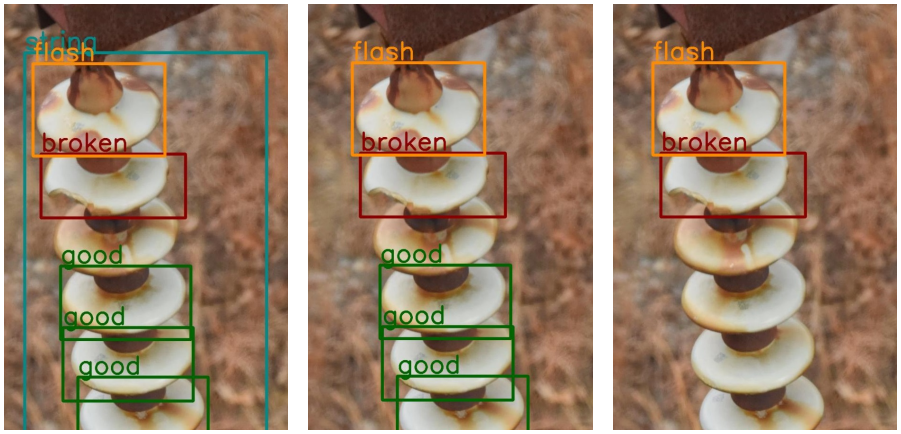
Object Shape	**Large Box**	**Small Box**	**Polygon**
Visual Sample			

Table 4.4: Methodology description of Categorical Definition

Classes Droped	**None [21]**	**String**	**String Good**
Visual Sample			

4.2.2 Results and Discussion

To explore how annotation consistency, spatial granularity, and categorical definition affect model performance, this section presents the comprehensive experimental computed results in a compact format, using custom and flexible plotting functions that combine multiple dimensions to effectively distinguish between stochastic variance and deterministic trends. From an inspection perspective, the focus lies on the model's ability to detect defective instances. Therefore, in the categorical definition experiments, the test set includes only defective classes (*broken* and *flash*), even though some models were trained with additional functional asset classes.

The results of the annotation consistency experiments indicate marginal variance for the *string* class, since it is a functional asset where the QC threshold is not applicable. It is also notable that the poor results for the *broken* class (AP50=49.3%) occur in the reference dataset [98], which aligns with its lower instance count (265). The standout data points are undoubtedly for the *flash* class, where the highest results (AP50=82.82%) also reflect the larger number of instances (889) annotated using a *careful* labeling policy. This conclusion confirms that overlooking annotation procedures, which leads to mislabeled data, limits the model's learning potential.

The spatial granularity experiments reveal an increasing trend in model performance, as the labeled objects are better fitted to their corresponding defective visual features. This highlights the importance of refining spatial annotations to enhance defect detection capabilities.

The categorical definition experiments exhibit considerable variance across both model and X-Val fold dimensions. However, no significant trends were observed to establish a correlation between the labeling strategy, specifically the inclusion of additional functional classes, and performance. This suggests that the presence of annotated functional assets does not necessarily improve the detection of defective ones.

Larger models have a greater capacity for learning complex patterns, making them suitable for tasks requiring detailed feature extraction. However, their tendency to memorize noise or irrelevant details makes them prone to overfitting, especially when datasets contain mislabeled data. This helps explain why larger model variants do not always outperform smaller ones.

4.2.3 Conclusion

This study evaluated the impact of various factors on model performance in object detection. Key dimensions such as annotation consistency, spatial granularity, and categorical definition were explored to understand their effects on the performance of trained models. Empirical results using a dataset of insulator images identified how different QC policies and labeling strategies influenced model performance metrics. The results suggest that annotation consistency and the granularity of object representation influences significantly model accuracy. Furthermore, while adding functional asset classes to the dataset did not significantly improve performance, variations in the QC policy and the number of instances annotated in the test set had a notable impact on the outcomes. One of the key insights from this study is the critical importance of data quality in model training, particularly regarding the annotation procedures applied to the training data. The findings reinforce the need for careful attention to QC procedures to avoid overfitting to noise. Explore contrastive learning loss functions [147], to leverage the inherent differences between defective and functional categories more effectively, and further investigation into the impact of annotation granularity could yield more robust results.

Chapter 5

Conclusion

This chapter concludes the proposed research, providing a succinct overview of the objectives, anticipated outcomes, and the potential impact of the project. It reflects on the key challenges and risks identified throughout the work and discusses the strategies to address these issues. The chapter also revisits the contributions of the research, with an emphasis on how it advances the field of electrical grid asset management through automated visual inspections. Additionally, potential risk management strategies are presented to ensure successful project execution.

5.1 Summary

The research proposed in this doctoral studies plan aims to advance automated visual inspection techniques for electrical grid assets by developing novel deep learning models, with a strong emphasis on data quality and multimodal learning. The integration of these models into the operational workflow of DSOs like EDP Labelec will not only enhance inspection efficiency but also ensure the reliability and safety of electrical grid infrastructure. The work will focus on addressing the limitations of current CV models, which often struggle with insufficient data and domain-specific challenges. The collaboration with EDP Labelec provides access to valuable real-world datasets, including images from UAVs equipped with high-resolution RGB, infrared, and LiDAR sensors. This partnership will also facilitate real-world validation of the developed models and their integration into existing workflows. Key objectives include the creation of a comprehensive, high-quality dataset for training deep learning models, the design of multimodal models for improved fault detection, and the development of lightweight models optimized for edge processing in resource-constrained environments. Through the use of active learning, the project aims to enhance data quality and model performance with minimal manual intervention. The proposed research will be structured around several core tasks, including data collection and preparation, deep learning model design, dataset annotation, and field testing. By addressing these tasks systematically, the project aims to develop a scalable, efficient solution for automated grid asset inspection. Additionally, the integration of these solutions into operational workflows will drive tangible benefits for the electrical grid industry, including reduced operational costs and improved maintenance efficiency. Ultimately, this research seeks to contribute to the growing field of data-centric AI by demonstrating the importance of high-quality, domain-specific data in achieving reliable, real-world performance in automated visual inspection tasks. The successful development and deployment of these models will represent a significant step toward the automation of asset inspection in electrical grids, improving both the operational effectiveness of DSOs and the overall reliability of energy distribution systems.

5.2 Risk Management

The research in deep learning (DL), particularly for visual inspection of electrical grid assets, faces multiple challenges that could influence its progress and outcomes. These challenges are inherent in the dynamic nature of scientific research, especially in the areas of AI and DL, and its application in real-world scenarios. The following outlines the key challenges and corresponding mitigation strategies, as well as alternative approaches to ensure the adaptability and success of the project:

- **Rapid Technological Change:** The fast-paced development of DL techniques can make it difficult to keep up with state-of-the-art methods. To mitigate this, continuous engagement with the latest literature, conferences, and collaborative networks will be essential.
- **Excessive Training Time:** Training deep learning models on large datasets can be time-consuming. A data sampling strategy, using a subset of the data, can speed up the training process while ensuring the model learns essential features.
- **Overwhelming Number of Tasks:** Research projects often become complex due to many tasks. Prioritizing critical tasks in early phases ensures focus, with secondary tasks deferred for later stages.
- **Resource Allocation:** Given the potential workload, some tasks may be delegated to master's students for isolated complementary experiments, allowing the principal researcher to focus on key aspects of the project.
- **Perfectionism and Scope Expansion:** Striving for perfection can delay progress. Balancing accuracy and practicality, using preliminary results to draw conclusions, will ensure continuous progress with room for refinement.
- **Balancing Breadth and Depth:** Exploring many aspects of a rapidly evolving field can lead to scattered results. The research will focus on specific objectives, ensuring thorough exploration and maintaining depth.
- **Publication Pressure:** PhD researchers often face pressure to publish quickly. The research will prioritize quality over quantity, ensuring publications are well-supported by experimentation and analysis.
- **Evaluation Metrics:** Defining success requires carefully selected evaluation criteria, considering real-world complexities. A range of metrics, including domain-specific ones, will evaluate the models' performance in controlled and real-world environments.
- **Interpretable and Explainable AI:** Many DL models are "black boxes." To ensure transparency, the research will use explainability techniques (e.g., SHAP, LIME) to clarify model decisions, ensuring trust in the system.
- **Ethical and Privacy Concerns:** Real-world data can contain sensitive information. Compliance with privacy laws like GDPR, alongside data anonymization and secure handling procedures, will mitigate privacy concerns.

These challenges provide a realistic view of the research landscape. By addressing them proactively, the proposed work will ensure steady progress and valuable contributions to deep learning applications in automated visual inspection. The alternative strategies outlined will maintain the research's flexibility, allowing for the effective resolution of obstacles without compromising the core objectives.

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